

Evaluating Classical Deepfake Detection Paradigms on the DF40 Image Benchmark

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Abstract—Recent years have seen tremendous advances in generative AI, making it more difficult to distinguish between artificially generated and real images. In response, deepfake detection has become a crucial field of study, and numerous models are currently performing well on common academic benchmarks. However, a significant drawback is that the majority of these models were tested and trained on datasets with a limited number of forgery techniques, calling into question how well they truly generalize. DF40 addresses this limitation by offering a large-scale benchmark with 40 different deepfake generation algorithms, ranging from GANs to more modern diffusion-based models and commercial tools. However, traditional detection methods have not been thoroughly tested on it. This work assesses three popular detection paradigms — MesoNet, EfficientNet-B4, and F3Net — each pre-trained on FaceForensics++, using the Entire Face Synthesis (EFS) and Face Editing (FE) image subsets of DF40 — the only categories provided in image format, as FS and FR are distributed exclusively as video. Our findings show that EfficientNet-B4 achieves the most consistent performance across both subsets (average AUC of 0.6952 on EFS and 0.6220 on FE), while F3Net degrades significantly on face editing techniques (AUC drop from 0.7149 to 0.5870), and MesoNet fails to generalize, exhibiting near-random AUC scores across most datasets. MesoNet underperforms due to an inherent bias favoring the fake category, whereas F3Net’s lower performance on FE indicates that the FAD branch, as implemented in DeepfakeBench, may struggle to generalize to face editing manipulations; however, this observation is based on an evaluation of that single branch only. The results hint at the need for more flexible and reliable detection approaches, suggesting that traditional detectors find it difficult to generalize to the entire range of modern forgery techniques.

Index Terms—deepfake detection, image forgery, generalization, FaceForensics++, DF40, MesoNet, EfficientNet-B4, F3Net, diffusion models

I. INTRODUCTION

Generative Artificial Intelligence (Gen AI) has advanced rapidly in recent years, making the creation of deepfakes easier than ever before [1]. This represents a growing threat across multiple domains, including misinformation, reputational damage, and privacy violations [2].

The most well-known deepfake detectors are F3Net [3], which mines forgeries in the frequency domain, EfficientNet-B4 [4], which uses transfer learning for binary classification of altered photos, and MesoNet [5], a lightweight CNN that targets mesoscopic characteristics of facial images. These models

are mostly trained and evaluated using FaceForensics++ [6], a benchmark built around only four forgery techniques, raising concerns about their capacity to generalize to more diverse manipulation methods.

DF40 [7] addresses this limitation by offering a comprehensive benchmark that includes 40 different deepfake generation methods, from classical GANs to recent diffusion-based models and commercial tools. Its image subset, which includes Face Editing (FE) and Entire Face Synthesis (EFS) techniques, provides a challenging evaluation setting that has not yet been studied using classical detection paradigms. To the best of our knowledge, this is the first work to assess MesoNet, EfficientNet-B4, and F3Net on the DF40 image subset using pre-trained weights without fine-tuning. While DF40 provides evaluations of eight detection methods [7], the specific combination of MesoNet, EfficientNet-B4, and F3Net evaluated in a zero-shot setting on the EFS and FE image subsets has not been reported.

To assess the performance of the evaluated detectors, we investigate two research questions: (RQ1) Do performance differences vary across forgery categories (EFS vs. FE), and if so, which architectural paradigm is most robust to each? (RQ2) Which specific DF40 generation methods cause the largest performance degradation? We hypothesize that models trained exclusively on FF++ will exhibit significant generalization gaps when evaluated on diverse modern forgery techniques, particularly those generated by diffusion-based and GAN-based methods not represented in their training data.

The rest of the paper is organized as follows: Section II reviews related work, Section III describes the methodology, Section IV presents the results, and Section V concludes the paper.

II. RELATED WORK

Detectors based on the spatial domain examine pixel-level data in the RGB color space to identify visual anomalies and artifacts produced during the forgery process [6]. Xception [6], a deep CNN based on depthwise separable convolutions, is one of the most widely used spatial detectors and a typical baseline in many benchmarks. MesoNet [5] uses a small network with few layers to focus on mesoscopic image properties, providing an efficient alternative for real-time detection. EfficientNet-B4

[4] is another well-known spatial detector frequently used as a baseline in comprehensive deepfake detection benchmarks. Even in controlled environments, spatial detectors are highly susceptible to video compression. Standard codecs like H.264 frequently eliminate the subtle manipulation traces these models rely on, significantly reducing detection accuracy on low-quality or compressed media.

This limitation is addressed by frequency-domain analysis, which reveals forgery artifacts that are detectable even under heavy compression, as well as unusual spectral patterns and local frequency statistics that are invisible to the human eye but clearly anomalous when examined using frequency transformations. F3Net [3] performs well on compressed media by extracting forgery patterns using two complementary frequency-aware clues: Frequency-aware Image Decomposition (FAD) and Local Frequency Statistics (LFS). Two other notable detectors in this area are SRM [8] and SPSL [9], which show robustness in cross-dataset evaluations by employing high-frequency noise characteristics and phase spectrum analysis, respectively.

Deepfake benchmark collections have grown in size and diversity, replacing smaller datasets. With approximately over 1.8 million manipulated images generated using four different methods, FaceForensics++ [6] has long been the primary training and evaluation benchmark. However, its limited diversity in both forgery methods and source conditions motivated the development of more difficult datasets, such as Celeb-DF [10] and the Deepfake Detection Challenge [11]. More recently, DF40 [7] addressed the lack of diversity and realism in earlier benchmarks by including 40 distinct deepfake generation approaches, including cutting-edge diffusion models. In spite of this progress, classical detection paradigms have yet to be tested on DF40, which is the gap that this study covers.

III. METHODOLOGY

We only used the DF40 [7] image subset for evaluation, concentrating on the Entire Face Synthesis (EFS) and Face Editing (FE) categories. A total of 456,587 images made up the evaluation set, which included 119,071 real images and 337,516 fake images (306,581 from EFS and 30,935 from FE) provided by the dataset authors. We observed that the StyleGAN2_ff and StyleGAN3_ff splits in the DF40 release contain the same image files, which results in identical metrics across all evaluated models. Nine corrupted images (unable to load via OpenCV) were excluded from evaluation across MidJourney and starganv2. The dataset authors had already applied face cropping to all images, allowing for direct model input. Due to computational constraints, this evaluation is limited to the EFS and FE image subsets of DF40, excluding face-swapping (FS) and face-reenactment (FR) categories, which are provided in video format and would require additional preprocessing. Additionally, the _ff and _cdf domain splits rely on different real-image datasets—FaceForensics++ and Celeb-DF, respectively—which can lead to discrepancies in resolution, compression, and face-cropping. These factors may

affect detector performance on their own, separate from the specific forgery technique being evaluated.

The three chosen models — MesoNet [5], EfficientNet-B4 [4], and F3Net [3] — were evaluated using the FaceForensics++ [6] pre-trained weights made available in the DeepfakeBench [12] repository, without any additional fine-tuning. MesoNet operates on face-cropped images and outputs a binary real/fake decision by analyzing mesoscopic-level features. EfficientNet-B4 uses the same type of input but performs classification through a deep spatial feature extraction pipeline. F3Net applies FAD to divide each image into frequency bands and LFS to examine local frequency statistics before creating its classification result. Although F3Net originally comprises two branches (FAD and LFS), the DeepfakeBench implementation uses only the FAD branch due to a known issue with the LFS implementation. A classification threshold of 0.5 was applied uniformly across all models, where predictions above this value were classified as fake and below as real. This zero-shot setting enables us to directly evaluate the generalization capability of each paradigm when confronted with previously unseen forgery techniques.

Three metrics were used to evaluate each model: AUC as the primary metric, consistent with DF40 [7] and DeepfakeBench [12], along with Accuracy and F1-score as complementary measures. Minor technical adaptations were required for compatibility, including nonstandard filename formats, path separator normalization, and GIF support for MidJourney subset. The evaluation set is imbalanced, with fake images outnumbering real images (337,516 vs. 119,071). Since no training is performed, this imbalance does not affect model weights. However, it may influence F1-score interpretation, which is why AUC is used as the primary evaluation metric, as it is threshold-independent and robust to class imbalance. All experiments were performed on a Dell Precision 7680 (Intel i9-13950HX, NVIDIA RTX 4000 Ada, 64 GB DDR5 RAM).

IV. RESULTS

Tables I through VI report the per-dataset performance of each model on the EFS and FE subsets of DF40. Results are presented separately for each paradigm to facilitate detailed analysis across forgery techniques.

A. *EfficientNet-B4*

In both subsets, EfficientNet-B4 performed the most consistently. On EFS (Table I), whichisreal produced the lowest AUC (0.4574), indicating challenges with GAN-based synthesis, whereas sd2.1_cdf produced the best AUC (0.9355). Performance was still moderate on FE (Table II), with starganv2 having the lowest AUC (0.4299) and e4e_ff having the highest (0.7391).

B. *F3Net*

F3Net showed strong EFS performance (Table III), with sd2.1_ff achieving an AUC of 0.9563. However, results degraded notably on FE (Table IV), particularly on starganv2 (AUC 0.3899) and styleclip (AUC 0.5605), indicating limited generalization to face editing techniques.

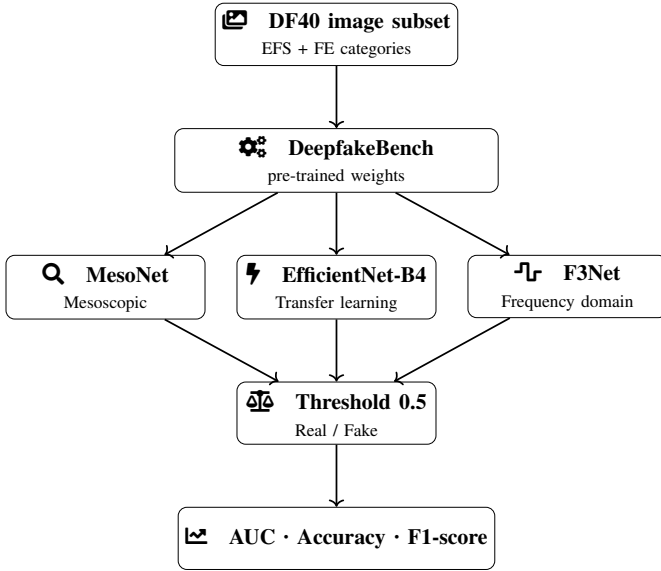


Fig. 1. Evaluation pipeline for assessing MesoNet, EfficientNet-B4, and F3Net on the DF40 image subset using pre-trained weights from DeepfakeBench.

TABLE I
EFFICIENTNET-B4 PERFORMANCE ON DF40 EFS SUBSET

Dataset	Accuracy	AUC	F1
StyleGAN2_ff	0.5886	0.5638	0.4170
StyleGAN3_ff	0.5886	0.5638	0.4170
StyleGANXL_ff	0.7740	0.8663	0.7463
sd2.1_ff	0.8433	0.9253	0.8369
ddim_ff	0.6971	0.7635	0.6279
DiT_ff	0.5194	0.5038	0.2448
SiT_ff	0.5438	0.5541	0.3097
pixart_ff	0.7206	0.7815	0.6665
rddm_ff	0.8797	0.8745	0.8793
VQGAN_ff	0.6547	0.7402	0.5525
MidJourney	0.5285	0.5610	0.4513
whichisreal	0.4823	0.4574	0.4066
StyleGAN2_cdf	0.5373	0.6419	0.6445
StyleGAN3_cdf	0.4421	0.5450	0.5375
StyleGANXL_cdf	0.5709	0.7006	0.6786
sd2.1_cdf	0.9067	0.9355	0.9442
ddim_cdf	0.5887	0.6796	0.6959
DiT_cdf	0.3673	0.5310	0.4408
SiT_cdf	0.4364	0.5834	0.5306
pixart_cdf	0.7335	0.8005	0.8221
rddm_cdf	0.9167	0.8943	0.9449
VQGAN_cdf	0.5713	0.6959	0.6790

StyleGAN2_ff and StyleGAN3_ff yield identical metrics; see Section III for details.

TABLE II
EFFICIENTNET-B4 RESULTS ON DF40 FE SUBSET

Dataset	Accuracy	AUC	F1
stargan	0.5474	0.5742	0.4210
starganv2	0.4825	0.4299	0.3088
styleclip	0.5991	0.6871	0.6537
e4e_ff	0.6657	0.7391	0.6105
e4e_cdf	0.5680	0.6566	0.6516
CollabDiff	0.6480	0.6453	0.5633

TABLE III
F3NET PERFORMANCE ON DF40 EFS SUBSET

Dataset	Accuracy	AUC	F1
StyleGAN2_ff	0.7128	0.7676	0.6220
StyleGAN3_ff	0.7128	0.7676	0.6220
StyleGANXL_ff	0.7695	0.8655	0.7176
sd2.1_ff	0.8916	0.9563	0.8845
ddim_ff	0.6574	0.7542	0.5135
DiT_ff	0.5363	0.5427	0.2048
SiT_ff	0.5540	0.5872	0.2577
pixart_ff	0.7683	0.8312	0.7157
rddm_ff	0.4775	0.7407	0.0037
VQGAN_ff	0.6466	0.7730	0.4905
MidJourney	0.7597	0.8054	0.7479
whichisreal	0.4473	0.3845	0.2266
StyleGAN2_cdf	0.5858	0.7230	0.6863
StyleGAN3_cdf	0.5133	0.6696	0.6099
StyleGANXL_cdf	0.4954	0.7220	0.5897
sd2.1_cdf	0.9163	0.9468	0.9493
ddim_cdf	0.4620	0.6469	0.5504
DiT_cdf	0.2666	0.4991	0.2674
SiT_cdf	0.3153	0.5491	0.3477
pixart_cdf	0.6969	0.7938	0.7882
rddm_cdf	0.2288	0.7754	0.0012
VQGAN_cdf	0.3955	0.6793	0.4650

StyleGAN2_ff and StyleGAN3_ff yield identical metrics; see Section III for details.

TABLE IV
F3NET RESULTS ON DF40 FE SUBSET

Dataset	Accuracy	AUC	F1
stargan	0.5930	0.6731	0.4176
starganv2	0.4832	0.3899	0.0777
styleclip	0.4974	0.5605	0.5183
e4e_ff	0.6249	0.7168	0.5022
e4e_cdf	0.5300	0.6590	0.5950
CollabDiff	0.5865	0.5228	0.4122

C. MesoNet

Overall, MesoNet had the weakest discrimination. Accuracy stayed close to 0.45 across almost all EFS datasets (Table V), with F1 scores artificially inflated by class imbalance — for example, 0.91 on cdf datasets — rather than reflecting meaningful detection, as confirmed by AUC values near 0.5. The FE data (Table VI) showed a similar pattern, where AUC values were typically close to chance. This suggests a model-level bias toward the fake class rather than a dataset imbalance effect, since EfficientNet-B4 and F3Net do not exhibit the same behavior under identical evaluation conditions.

V. CONCLUSION

This study employed pre-trained weights from FaceForensics++ without fine-tuning to evaluate three classic deepfake detection paradigms on the DF40 image subset: MesoNet, EfficientNet-B4, and F3Net. When tested against the various forgery techniques found in DF40, the findings revealed generalization limitations across all three models in this zero-shot setting, as shown in Table VII.

In response to RQ1, performance differences across EFS and FE varied substantially by paradigm. EfficientNet-B4

TABLE V
MESONET PERFORMANCE ON DF40 EFS SUBSET

Dataset	Accuracy	AUC	F1
StyleGAN2_ff	0.4523	0.5029	0.6228
StyleGAN3_ff	0.4523	0.5029	0.6228
StyleGANXL_ff	0.4531	0.4351	0.6236
sd2.1_ff	0.4531	0.5731	0.6236
ddim_ff	0.4530	0.4830	0.6235
DiT_ff	0.4531	0.4850	0.6236
SiT_ff	0.4531	0.5113	0.6236
pixart_ff	0.4531	0.7171	0.6236
rddm_ff	0.4531	0.6701	0.6236
VQGAN_ff	0.4531	0.5055	0.6236
MidJourney	0.4688	0.3527	0.6376
whichisreal	0.5017	0.5679	0.6673
StyleGAN2_cdf	0.8336	0.5019	0.9093
StyleGAN3_cdf	0.8337	0.5536	0.9093
StyleGANXL_cdf	0.8337	0.4627	0.9093
sd2.1_cdf	0.8337	0.5846	0.9093
ddim_cdf	0.8336	0.5703	0.9092
DiT_cdf	0.8337	0.4786	0.9093
SiT_cdf	0.8337	0.5079	0.9093
pixart_cdf	0.8337	0.7124	0.9093
rddm_cdf	0.7140	0.5196	0.8331
VQGAN_cdf	0.8337	0.5082	0.9093

StyleGAN2_ff and StyleGAN3_ff yield identical metrics; see Section III for details.

TABLE VI
MESONET RESULTS ON DF40 FE SUBSET

Dataset	Accuracy	AUC	F1
stargan	0.4995	0.5100	0.6662
starganv2	0.5010	0.4695	0.6667
styleclip	0.7390	0.5765	0.8497
e4e_ff	0.5000	0.4581	0.6667
e4e_cdf	0.7684	0.4445	0.8690
CollabDiff	0.5000	0.6610	0.6662

TABLE VII
PERFORMANCE COMPARISON ON DF40 IMAGE SUBSETS

Model / Subset	Accuracy	AUC	F1
EfficientNet-B4	0.6227	0.6789	0.6024
EFS	0.6335	0.6952	0.6218
FE	0.5851	0.6220	0.5348
F3Net	0.5708	0.6865	0.4875
EFS	0.5760	0.7149	0.5066
FE	0.5525	0.5870	0.4205
MesoNet	0.6212	0.5305	0.7525
EFS	0.6316	0.5335	0.7587
FE	0.5847	0.5199	0.7308

Unweighted mean per subset; each dataset weighted equally. StyleGAN3_ff excluded from means due to identical content with StyleGAN2_ff.

had the strongest performance, achieving the highest average AUC on both the EFS (0.6952) and FE (0.6220) subsets, with only a 7.32 percentage point difference between them. F3Net performed well on EFS (AUC 0.7149) but poorly on FE (AUC 0.5870), though these results reflect only the FAD branch as implemented in DeepfakeBench and may not fully represent the original two-branch architecture. MesoNet failed to yield meaningful predictions on the majority of datasets,

with predictions mismatching the ground truth labels and AUC values near random chance.

In response to RQ2, GAN-based methods (whichisreal, starganv2) and transformer-based diffusion (DiT, SiT) consistently produced the largest performance degradation across all detectors.

These results confirm that conventional detectors trained on FF++ struggle to generalize to contemporary generative approaches. Future work should investigate fine-tuning strategies on diverse datasets, domain adaptation techniques, and ensemble approaches combining spatial and frequency-domain information. Extending this evaluation to the FS and FR video subsets of DF40 would also provide a more comprehensive assessment of generalization across all forgery categories.

AI USAGE STATEMENT

The authors used Claude (Anthropic) as an AI writing assistant to support drafting, academic phrasing, and structural guidance. All content, arguments, analysis, and experimental results represent the authors’ own work and judgment.

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